

UA-FER: Uncertainty-aware representation learning for facial expression recognition

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ABSTRACT

Facial Expression Recognition (FER) remains a challenging task due to unconstrained conditions like variations in illumination, pose, and occlusion. Current FER approaches mainly focus on learning discriminative features through local attention and global perception of visual encoders, while neglecting the rich semantic information in the text modality. Additionally, these methods rely solely on the softmax-based activation layer for predictions, resulting in overconfident decision-making that hampers the effective handling of uncertain samples and relationships. Such insufficient representations and overconfident predictions degrade recognition performance, particularly in unconstrained scenarios. To tackle these issues, we propose an end-to-end FER framework called UA-FER, which integrates vision-language pre-training (VLP) models with evidential deep learning (EDL) theory to enhance recognition accuracy and robustness. Specifically, to identify multi-grained discriminative regions, we propose the Multi-granularity Feature Decoupling (MFD) module, which decouples global and local facial representations based on image-text affinity while distilling the universal knowledge from the pre-trained VLP models. Additionally, to mitigate misjudgments in uncertain visual-textual relationships, we introduce the Relation Uncertainty Calibration (RUC) module, which corrects these uncertainties using EDL theory. In this way, the model enhances its ability to capture emotion-related discriminative representations and tackle uncertain relationships, thereby improving overall recognition accuracy and robustness. Extensive experiments on in-the-wild and in-the-lab datasets demonstrate that our UA-FER outperforms the state-of-the-art models.

1. Introduction

Facial expressions are intuitive visual signals that convey human emotions, rich in subtle emotional content. Facial Expression Recognition (FER), as a critical topic in the computer vision community, aims to automatically recognize facial expressions from images. With advances in affective computing, FER has gained significant attention due to its widespread applications in various fields, such as human-computer interaction, mental health assessment, and autonomous driving [1].

In recent decades, numerous efforts have been made, from hand-crafted features to deep learning methods. Early approaches mainly rely on hand-crafted features, such as LBP [2], HOG [3], and SIFT [4], achieving promising performance on in-the-lab datasets. With the advancement of deep learning techniques, large-scale in-the-wild datasets, e.g., AffectNet [5], RAF-DB [6], and ExpW [7] have been proposed, which are more challenging to recognize. In these in-the-wild FER datasets, facial expression images are sampled from various complex scenes, leading to variations in head posture, illumination, and occlusion. Despite remarkable progress, FER remains unique challenges

distinct from general image classification tasks, primarily due to difficulties in two aspects. On the one hand, the discrepancies among different expressions are very subtle, termed high *inter-class similarities*. It is difficult to distinguish between emotion categories because muscle movements are similar and contained in small facial parts such as eyebrows, noses, and chins [8]. On the other hand, the same expression can exhibit diverse appearances, termed high *intra-class variabilities*. Expressions within the same class can exhibit strikingly different appearances, influenced by various factors such as race, gender, age, and cultural background [9].

To overcome these challenges, contemporary deep learning methods utilize local attention and global perception mechanisms to extract discriminative facial representations, as illustrated in Fig. 1(a). Specifically, local attention-based methods [10–15] adaptively focus on specific facial regions. In contrast, global perception-based methods [9,16–18] aim to capture global information from facial images to improve FER recognition performance. Despite substantial progress,

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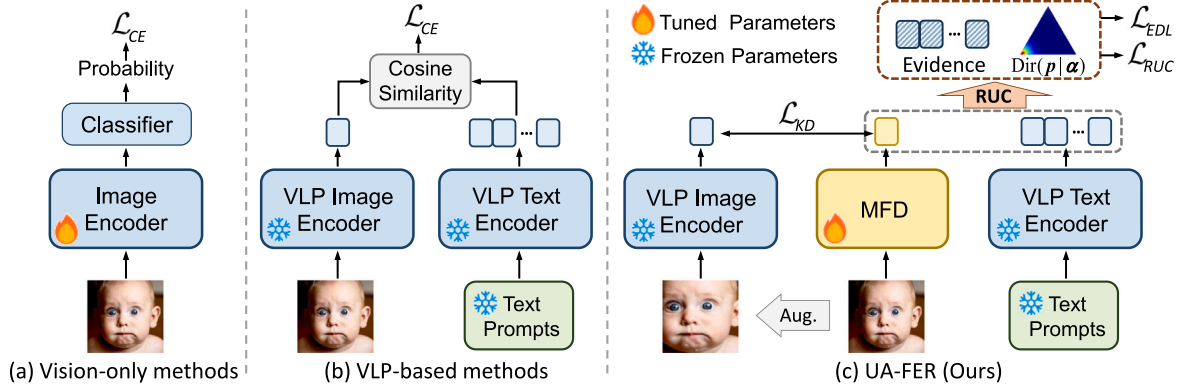


Fig. 1. Illustration of (a) Vision-only methods, (b) VLP-based methods, and (c) our proposed UA-FER. Compared to the other two type of methods, UA-FER not only incorporates the text modality into FER but also corrects the uncertain visual-textual relationships, improving recognition performance eventually.

both approaches exhibit two key deficiencies: (1) **Insufficient Representation**: Most existing FER methods primarily depend solely on visual encoders for feature extraction, neglecting the potential semantic value of the text modality. Text can provide rich contextual information that complements visual representations. For example, textual descriptions of emotions (e.g., “a boy with an angry expression”) provide semantic cues to capture subtle emotional expressions, enhancing the model’s representation capacity. (2) **Overconfident Determination**: Current FER methods predominantly rely on softmax-based classification, which outputs a single probability distribution across classes while ignoring uncertainty in predictions. These overconfident predictions indicate the model is “pretending to know”, even when lacking a true understanding of facial expressions. Specifically, expression categories in FER exhibit subtle differences (e.g., both “fear” and “surprise” feature an open mouth movement), frequently lying near decision boundaries. In such cases, softmax directly predicts the expression according to the probability distribution, even when other candidate categories have high probabilities, failing to account for ambiguity or uncertainty in decision-making.

Recently, vision-language pre-training (VLP) models, such as CLIP [19], have demonstrated strong cross-modal learning capabilities guided by text, as illustrated in Fig. 1(b). Building on the success of CLIP-like VLP models, several follow-up methods have been applied to various downstream tasks using pre-trained VLP models, such as image classification [20,21], object detection [22], and semantic segmentation [23]. Inspired by this, several studies have explored applying CLIP in FER task [24,25]. For example, CLIPER [24] employs the pre-trained CLIP model to learn facial expression representations, while CEPrompt [25] proposes learnable cross-modal prompts to enhance the interaction between vision and text modalities. However, these approaches still rely on softmax-based deterministic predictions, failing to account for uncertainty adequately. This oversight may lead to decreased recognition performance, especially when dealing with ambiguous or complex visual-textual relationships. Introducing uncertainty estimation has been shown to effectively prevent models from making incorrect decisions in uncertain conditions, thereby enhancing identification reliability [26]. Traditional methods, e.g., Dropout [27], estimate uncertainty through multiple forward passes but are often computationally expensive. In contrast, Evidential Deep Learning (EDL) [28] requires only a single forward pass to estimate uncertainty and has demonstrated potential across various tasks [29–31].

To address the above issues, we propose an end-to-end uncertainty-aware representation learning framework for FER, called UA-FER, which integrates the pre-trained CLIP model with EDL theory, as illustrated in Fig. 1(c). UA-FER fully leverages both vision and text modalities while preserving the integrity of image-text relationships, thereby enhancing the accuracy and robustness of FER model. Specifically, the

pre-trained CLIP model serves as the backbone network, extracting both visual and textual information. To capture multi-grained discriminative features in facial expressions, we propose the **Multi-granularity Feature Decoupling (MFD)**, which decouples global facial features and local features with high visual-textual affinity, corresponding to emotion-related discriminative regions. Additionally, we propose the **Universal Knowledge Retention (UKR)** module to efficiently distill generalized knowledge from the pre-trained CLIP model. However, the generated visual-textual affinity matrices might contain ambiguous or uncertain relationships, potentially leading to incorrect model predictions. To mitigate misjudgments on uncertain samples or relationships, we introduce the **Relation Uncertainty Calibration (RUC)** module, which models and corrects uncertainties in visual-textual relationships using EDL theory. Specifically, we appeal the model to train towards “Accurate and Certain” and “Inaccurate and Uncertain” outcomes by optimizing an objective that minimizes scenarios where predictions are “Accurate but Uncertain” or “Inaccurate but Certain”, thereby significantly enhancing recognition robustness and reliability.

The main contributions can be summarized as follows:

- (1) We propose a uncertainty-aware representation learning model for FER, namely UA-FER, which incorporates cross-modal information for representation learning based on the pre-trained CLIP model, while correcting the uncertain visual-textual relationships therein.
- (2) To identify the emotion-related facial variations, we propose a Multi-granularity Feature Decoupling (MFD) module, capturing discriminative and universal visual representations guided by the text. Besides, to alleviate the uncertain relationships arising from the visual-textual matching process, a Relation Uncertainty Calibration (RUC) strategy is introduced, enabling the model to better understand emotion-aware representations robustly.
- (3) Extensive experimental results on three in-the-wild and one in-the-lab FER benchmarks show the superiority of the proposed method.

2. Related work

In this section, related works about Facial Expression Recognition and Evidential Deep Learning are reviewed briefly.

2.1. Facial expression recognition

Facial Expression Recognition (FER) has garnered significant attention in recent decades due to its importance in various applications. Considerable progress has been made from traditional methods to deep learning techniques. Earlier traditional methods primarily relied on hand-crafted features such as LBP [2], HOG [3] and SIFT [4]. However,

these methods heavily depend on expert knowledge and struggle in complex scenarios such as lighting changes and occlusion.

With the rapid advancement of deep neural networks in various computer vision tasks, numerous deep learning-based FER methods have been introduced, which can be classified into two categories: local attention-based methods and global perception-based methods. Local attention-based methods [10–15,32–38] focus on specific facial regions and extract meaningful emotional representations. For example, Li et al. [39] identify the most valuable regions by dividing the facial image into small areas using facial landmarks and designing an attention module to learn significant relationships between these regions. Wang et al. [8] utilize an attention network to capture the salient features of each facial region. Ruan et al. [40] propose a feature decomposition and reconstruction network to recognize similarities and distinctive expression variations. Liu et al. [13] capture diverse and important emotional information by focusing on various regional features. Tao and Duan [36] propose a hierarchical attention network with progressive feature fusion for FER. Zhu et al. [38] design an emotion knowledge-based fine-grained network to capture inter-class relationships and discriminative representations of fine-grained emotions.

Recently, global perception-based methods [9,16–18,41] have also been explored, mainly using multi-head attention mechanisms [42] to enhance model performance. For example, Xue et al. [9] design an adaptive attention module for learning multiple distinct local patches, which is later combined to capture the relationships between these patches. Based on this, they process input tokens by reducing the number of tokens using a pooling scheme and then feeding the processed tokens into the ViT encoders [18]. Moreover, Mao et al. [41] combine facial images with landmarks for multi-scale feature extraction. This approach effectively reduces the complexity of the input while enhancing the model's ability to integrate global and local features, thereby increasing accuracy and robustness.

Although notable progress has been achieved, these approaches rely solely on visual encoders to extract features, overlooking the rich semantic information embedded in text. More recent studies have attempted to leverage pre-trained vision–language models [19] to extract information from both visual and textual modalities, leading to significant progress. For instance, Li et al. [24] employ the pre-trained CLIP model combined with multiple text descriptors to extract expression-related representations. However, they primarily extract global representations, overlooking fine-grained local facial details. Zhou et al. [25] employ learnable visual and textual prompts to enhance information interaction between visual and textual modalities.

However, these methods still rely on softmax-based prediction, disregarding the ambiguities and uncertainty in the image–text relationships. To address this issue, we introduce Evidence Theory Learning to model and correct interactions between visual and textual modalities, thereby further enhancing the recognition performance and robustness of the FER model.

2.2. Evidential deep learning

In recent decades, despite the great success of deep neural networks (DNNs) [19,25,43,44], most deep models use a softmax function as classification head for prediction. However, it fails to estimate the credibility of decision as the human decision-making process does. On the one hand, softmax-based classifiers tend to be overconfident in false predictions, complicating the optimization process [45]. On the other hand, since softmax outputs are essentially point estimates of probability distributions, they cannot estimate prediction uncertainty [30]. To address these issues, Evidential Deep Learning (EDL) [28] uses Subjective Logic (SL) theory [46] to construct a probabilistic framework where predictions are represented as belief distributions (*i.e.*, Dirichlet distributions) over the class probabilities. EDL initially requires the model to gather evidence for each class to establish confidence

in assigning inputs to specific categories. The collected evidence is subsequently used to parameterize a Dirichlet distribution, enabling the model to quantify predicted probabilities and their associated uncertainties. In this way, the model is trained to make more reliable and robust decisions, especially in challenging scenarios involving ambiguous samples. Subsequently, numerous studies refine the evidence collection processes from diverse aspects, including hyper-evidence allocation [47], sample evidence re-weighting [48], evidence function selection [49], and multi-level evidence collection [50]. Furthermore, some studies [51,52] adopt advanced evidence collection techniques and diverse evidence sources to replace the traditional reliance on model output logits.

Due to its strong usability and extensibility, EDL has also been widely applied to enhance existing classical machine learning paradigms in several computer vision tasks [26,29–31]. For example, Bao et al. [29] incorporate evidence learning to handle open-set action recognition. Chen et al. [30] leverage video-level uncertainty based on EDL theory to assess the impact of background noise on predictions for weakly-supervised temporal action localization. Qin et al. [31] and Huang et al. [53] enhance robust cross-modal retrieval by modeling uncertainty in cross-modal correspondence. In this paper, to the best of our knowledge, we are the first to incorporate evidential deep learning to model and correct the uncertainty between facial expressions and emotional text for facial expression recognition.

3. Proposed method

In this section, we first provide a brief revisit of Contrastive Language-Image Pre-training and Evidential Deep Learning. After that, we specify the proposed UA-FER, including Multi-granularity Feature Decoupling and Relation Uncertainty Calibration.

3.1. Preliminaries

3.1.1. Contrastive language-image pre-training

In recent years, inspired by the success of pre-trained models in the fields of computer vision [42,54] and natural language processing [55,56], vision–language pre-training (VLP) models [19,57–59] have received increasing attention in the CV community.

As a typical VLP model, Contrastive Language-Image Pre-training (CLIP) [19] model is trained with 400 million image–text pairs by vision–language contrastive learning to align the image and text embedding space. Specifically, CLIP model is composed of two crucial encoders: an image encoder $f_v(\cdot)$ and a text encoder $f_t(\cdot)$. Given an expression category $class_c$ ($c = 1, 2, \dots, C$), where C denotes the number of categories, a manually designed text prompt $\mathbf{p}_c$ is constructed as “a photo of a $[class_c]$ ”. The text encoder $f_t(\cdot)$ here is a Transformer [55] with architectural modifications as described in Radford et al. [60]. Following [19], we employ 12-layer Transformer blocks with 8 attention heads and cap the maximum sequence length of each input text at 76. Subsequently, the prompt is entered into $f_t(\cdot)$, obtaining a textual embedding $\mathbf{t}_c = g(\mathbf{p}_c)$. Simultaneously, an image $\mathbf{x}$ is fed into the image encoder $f_v(\cdot)$ to obtain the visual embedding $\mathbf{v} = f_v(\mathbf{x})$. The architecture of the image encoder $f_v(\cdot)$ in CLIP is generally based on CNNs like ResNet [44] or Vision Transformer (ViT) [42]. Following [24,25], we opt for ViT-B/16 as the image encoder with 12-layer Transformer blocks. Each block contains a multi-head self-attention module and a feed-forward network, enabling the model to capture rich visual representations by attending to global patch relationships across the entire image. Note that the image embeddings and the text embeddings output by the two encoders are linearly projected into the same multi-modal embedding space. After that, the prediction probability of $\mathbf{x}$ is formulated as:

$$p(y = c | \mathbf{x}) = \frac{\exp(\cos(\mathbf{v}, \mathbf{t}_c) / \tau)}{\sum_{i=1}^{|C|} \exp(\cos(\mathbf{v}, \mathbf{t}_i) / \tau)}, \quad (1)$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity, and τ is a temperature parameter.

3.1.2. Evidential deep learning

Existing deep learning models typically use softmax layers in deep neural networks for classification. However, these methods fail to estimate prediction uncertainty since the softmax score is a point estimate of the prediction distribution. Consequently, the outputs of networks tend to be overconfident in false predictions [45].

To address above limitations, Evidential Deep Learning (EDL) [28, 61] is proposed based on Dempster–Shafer theory [62] and the Subjective Logic theory [46]. Instead of directly predicting the probability of each category, EDL collects the evidence for each category and then constructs the Dirichlet distribution of class probabilities parametrized over the collected evidence. Based on this distribution, the multiclass classification and prediction uncertainty can be quantified jointly.

The target of EDL is to predict the evidence for each category and to establish the Dirichlet distribution of class probabilities. In particular, given a sample $\mathbf{x}$ for C -class classification, the evidence vector $\mathbf{e} = f_e(\mathbf{x}) \in \mathbb{R}_+^C$ can be predicted, where $f_e(\cdot)$ is the evidence extractor (e.g., relu) to ensure the evidence e non-negative. Subjective Logic tries to assign a belief mass and an overall uncertainty mass based on the collected evidence, which could be defined as:

$$b_c = \frac{e_c}{S} = \frac{\alpha_c - 1}{S} \quad \text{and} \quad u = \frac{C}{S}, \quad (2)$$

where $S = \sum_{c=1}^C \alpha_c = \sum_{c=1}^C (e_c + 1)$, and $u + \sum_{c=1}^C b_c = 1$. The S is regarded as the Dirichlet distribution strength. The corresponding Dirichlet distribution is given by:

$$\text{Dir}(\mathbf{p}|\boldsymbol{\alpha}) = \begin{cases} \frac{1}{B(\boldsymbol{\alpha})} \prod_{c=1}^C p_c^{\alpha_c-1}, & \text{for } \mathbf{p} \in S_C, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Here, $\alpha_c = e_c + 1$ are the parameters of $\text{Dir}(\mathbf{p}|\boldsymbol{\alpha})$, $B(\boldsymbol{\alpha})$ is a C -dimensional Beta function and S_C is a C -dimensional unit simplex. Treating $\text{Dir}(\mathbf{p}|\boldsymbol{\alpha})$ as the distribution of class probability, the EDL loss can be minimized for learning evidence e_c eventually as following:

$$\mathcal{L}_{\text{EDL}} = \sum_{c=1}^C y_c \cdot (\log(S) - \log(e_c + 1)), \quad (4)$$

where y is an one-hot ground-truth label for sample $\mathbf{x}$. In the inference phase, the predicted probability $\hat{p}_c$ of class c :

$$\hat{p}_c = \frac{\alpha_c}{S}. \quad (5)$$

3.2. UA-FER

The overview framework of the proposed UA-FER is shown in Fig. 2. Specifically, to capture emotion-related discriminative representations in facial images, we introduce the Multi-granularity Feature Decoupling (MFD) module, which focuses on multi-grained discriminative facial regions. Given a facial image $\mathbf{x}$, emotion-aware global and local representations are obtained by the Emotion-aware Adapter. Subsequently, we decouple these representations using affinity matrices to mitigate the effects of inter-class similarity and intra-class variability. Furthermore, we distill universal knowledge from the pre-trained CLIP to retain generalizable representations. Additionally, to minimize the model's misjudgment of uncertain samples, we propose the Relation Uncertainty Calibration (RUC) strategy based on EDL theory, enhancing the model's performance and reliability in recognition.

3.2.1. Multi-granularity feature decoupling

The CLIP model is widely applied to various downstream tasks due to its text-based pre-training, which demonstrates strong generalization ability. However, in facial expression recognition, the emotion-related fine-grained facial representations are neglected since it focuses on coarse-grained representations. To this end, we propose Multi-granularity Feature Decoupling (MFD), which decouples emotion-

aware features from facial content.

Emotion-aware Adapter. To extract discriminative emotion-aware representations from limited facial expression samples, we introduce the Emotion-aware Adapter, training it instead of fine-tuning the entire CLIP network. Specifically, given an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, where H and W are the height and width of $\mathbf{x}$, we partition it into N patches of spatial size $P \times P$. Then, a sequence of image embeddings $\mathbf{v} = \{\mathbf{v}_{cls}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\} = f_{EA}(\mathbf{x})$ is obtained after processing through sequentially stacked Transformer blocks. Here, $\mathbf{v}_{cls}$ denotes the class token embedding, while $\mathbf{v}_n (n = 1, 2, \dots, N)$ represents the local patch embedding.

Note that the Emotion-aware Adapter $f_{EA}(\cdot)$ shares the same structure as CLIP's image encoder $f_v(\cdot)$, i.e., a ViT network with 12-layer transformer blocks. The key distinction, however, lies in the utilization of parameters within the two networks. Specifically, as shown in Fig. 2, $f_{EA}(\cdot)$ is fine-tuned to extract emotion-specific representations, whereas $f_v(\cdot)$ retains frozen parameters to preserve the generalization capabilities learned during CLIP's pretraining (as described in Eq. (9)). This distinction allows the adapter to specialize in emotion recognition without compromising the universal knowledge provided by CLIP.

Affinity-based Feature Decoupling. As described in Section 3.1.1, given an expression category $class_c$ (e.g., "surprise"), an emotion text prompt such as "a photo of a [$class_c$] facial expression" is constructed and fed into the text encoder to generate a textual embedding $\mathbf{t}_c$. We then compute the affinity matrices between the visual embeddings $\mathbf{v}_{cls}$, $\mathbf{v}_n$ and textual embedding $\mathbf{t}_c$:

$$\begin{aligned} \mathcal{A}_{cls}^{(c)} &= \frac{\exp(\cos(\mathbf{v}_{cls}, \mathbf{t}_c)/\tau)}{\sum_{j=1}^C \exp(\cos(\mathbf{v}_{cls}, \mathbf{t}_j)/\tau)}, \\ \mathcal{A}_n^{(c)} &= \frac{\exp(\cos(\mathbf{v}_n, \mathbf{t}_c)/\tau)}{\sum_{j=1}^C \exp(\cos(\mathbf{v}_n, \mathbf{t}_j)/\tau)}, \end{aligned} \quad (6)$$

where $\mathcal{A}_{cls}^{(c)}$ denotes the global affinity, and $\mathcal{A}_n^{(c)} (n = 1, 2, \dots, N)$ refers the local affinity.

However, facial patches often contain redundant or emotion-irrelevant representations due to complex backgrounds or varying poses. To mitigate the impact of these irrelevant representations and maximize the focus on emotion-related representations, we decouple the local affinities $\mathcal{A}_{1:N}^{(c)}$ based on the computed affinity matrices. As illustrated in Fig. 2, the top K affinities from the affinity matrix are selected. These global and local affinities are combined to the weighted affinity $\mathbb{A}^{(c)}$:

$$\begin{aligned} \mathbb{A}^{(c)} &= \gamma \cdot \mathcal{A}_{cls}^{(c)} + (1 - \gamma) \cdot \mathcal{A}_{loc}^{(c)}, \\ \mathcal{A}_{loc}^{(c)} &= \text{Mean}(\text{TopK}(\mathcal{A}_1^{(c)}, \mathcal{A}_2^{(c)}, \dots, \mathcal{A}_N^{(c)})). \end{aligned} \quad (7)$$

Here, the global factor γ is used to balance the effects of global and local affinities, $\mathcal{A}_{loc}^{(c)}$ denotes the mean local affinity and $\text{TopK}(\cdot)$ is the top-k function. Then, $\mathbb{A}^{(c)}$ is converted into probabilities and optimized using standard cross-entropy (CE) loss as following form:

$$\begin{aligned} \mathcal{L}_{\text{CE}} &= -\frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \sum_{j=1}^C y_i^{(j)} \cdot \log(p_i^{(j)}), \\ p &= \text{softmax}(\mathbb{A}), \end{aligned} \quad (8)$$

where, $\mathcal{N}$ denotes the total number of samples.

Universal Knowledge Retention. To maintain the strong generalization capability of the pre-trained CLIP model, which inherently captures robust representations (e.g., content, structure, and other visual elements), we retain the CLIP's universal knowledge within the Adapter. Specifically, the universal class token embedding $\mathbf{v}_{clip} = f_v(\tilde{\mathbf{x}})$ is obtained, where $\tilde{\mathbf{x}} = DA(\mathbf{x})$ represents the data-augmented counterpart (e.g., random cropping and flipping). Here, the CLIP image encoder $f_v(\cdot)$ serves as the teacher network, while the Emotion-aware Adapter $f_{EA}(\cdot)$ functions as the student network. The knowledge distillation process is then formulated as follows:

$$\mathcal{L}_{\text{KD}} = \|\mathbf{v}_{clip} - \mathbf{v}_{cls}\|_1, \quad (9)$$

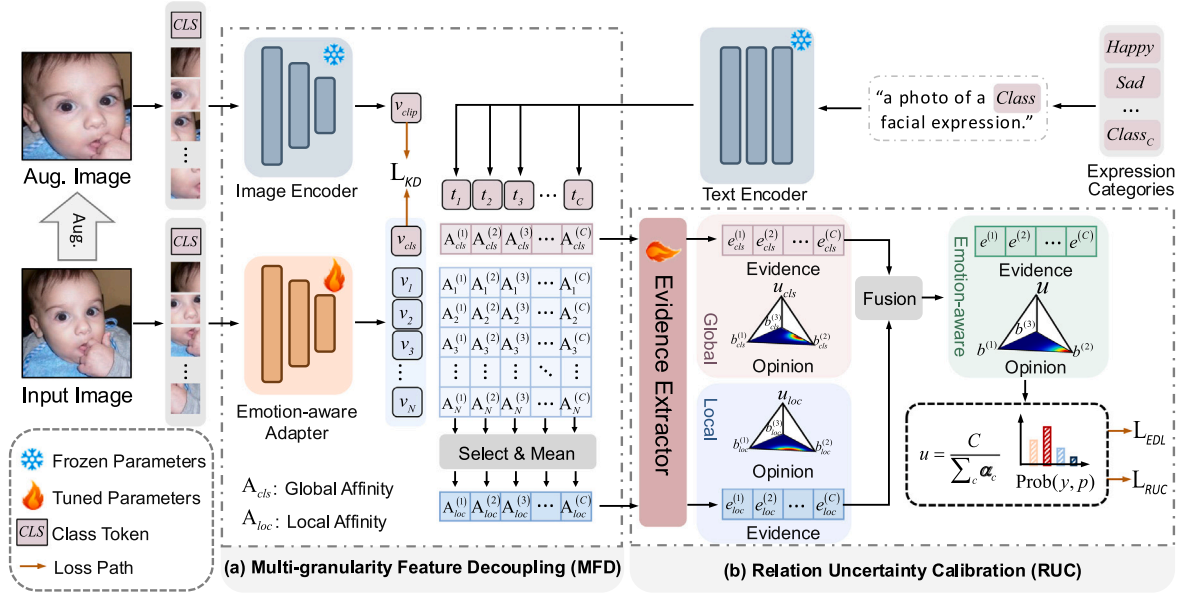


Fig. 2. The overview of our proposed UA-FER, which consists of two crucial modules: Multi-granularity Feature Decoupling (MFD) to extract emotion-aware representations, and Relation Uncertainty Calibration to improve the reliability during recognition.

where $\|\cdot\|_1$ refers to the L1 regularization. This ensures that while the Adapter is fine-tuned for emotion-aware tasks, it still leverages the generalized knowledge originally encoded by CLIP.

3.2.2. Relation uncertainty calibration

In the MFD module, the softmax-based prediction process is inadequate for estimating classification uncertainty, leading to overconfidence when making incorrect predictions. To address the above issue, we propose the Relation Uncertainty Calibration (RUC) strategy based on Evidential Theory Learning (EDL) theory during model optimization. In this way, the uncertainties in vision-language relations are corrected, enabling the model to better understand emotion-aware representations robustly.

Given the global affinity $\mathcal{A}_{cls}$ and local affinity $\mathcal{A}_{loc}$ (as described in Eq. (7)), which indicates the similarity between expression images and emotion text), the global and local evidence can be extracted using the evidence extractor $f_e(\cdot)$, defined as:

$$\begin{aligned} e_{cls} &= f_e(\mathcal{A}_{cls}) = [e_{cls}^{(1)}, e_{cls}^{(2)}, \dots, e_{cls}^{(C)}], \\ e_{loc} &= f_e(\mathcal{A}_{loc}) = [e_{loc}^{(1)}, e_{loc}^{(2)}, \dots, e_{loc}^{(C)}]. \end{aligned} \quad (10)$$

Taking the three-classification problem ($C=3$) as an example, as shown in Fig. 2(b), we can derive the belief mass $\{b_{cls}^{(c)}\}_{c=1}^C$ and uncertainty mass u_{cls} corresponding to the global opinion based on the given global evidence e_{cls} through Eq. (2). Likewise, the local belief mass $\{b_{loc}^{(c)}\}_{c=1}^C$ and uncertainty mass u_{loc} are obtained, based on the local evidence e_{loc} . We then fuse the above two types of evidence to obtain the emotion-aware evidence by:

$$e = \gamma \cdot e_{cls} + (1 - \gamma) \cdot e_{loc}. \quad (11)$$

Here, the balance factor γ is used to balance the global and local evidences as in Eq. (7). After that, as described in Eq. (2) to (3), the Dirichlet parameter $\alpha = [\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(C)}]$, belief mass $b = [b^{(1)}, b^{(2)}, \dots, b^{(C)}]$ and the uncertainty u can be obtained.

Following [26,29], we maximize the Accuracy versus Uncertainty (AvU) utility function to calibrate the relation uncertainty between facial images and emotion text. The estimation process of the AvU metric can be grouped into four cases: [1] Accurate and Certain (AC), [2] Accurate and Uncertain (AU), [3] Inaccurate and Certain (IC), and [4] Inaccurate and Uncertain (IU), as shown in Fig. 3.

$$\text{AvU} = \frac{\mathcal{N}_{AC} + \mathcal{N}_{IU}}{\mathcal{N}_{AC} + \mathcal{N}_{AU} + \mathcal{N}_{IC} + \mathcal{N}_{IU}}, \quad (12)$$

where the $\mathcal{N}_{AC}$, $\mathcal{N}_{AU}$, $\mathcal{N}_{IC}$, and $\mathcal{N}_{IU}$ represent the numbers of samples in the predicted cases AC, AU, IC and IU, respectively.

A reliable and well-calibrated model should achieve a higher AvU measure ($\text{AvU} \in [0,1]$), ensuring that the prediction uncertainty be consistent with the corresponding probability. To calibrate prediction uncertainty, we encourage the model to learn sharp Dirichlet simplexes (Fig. 3(a)) for correct predictions, ensuring high confidence in these cases. Conversely, for incorrect predictions, the model should learn smooth Dirichlet simplexes (Fig. 3(e)) to signal uncertainty. In other words, the model should maximize confidence in correct predictions while expressing greater uncertainty in incorrect ones. To this end, we propose standardizing the EDL training process by minimizing the expectation of the AU and IC cases (Fig. 3(c) and (d)). Thus, the model is forced to learn the other two extreme cases, i.e., AC in Fig. 3(a) and IU in Fig. 3(e), ultimately enhancing the accuracy and reliability of uncertainty predictions.

The proposed RUC calibrates the prediction uncertainty by minimizing the logarithmic constraint between confidence p and uncertainty u as follows:

$$\mathcal{L}_{RUC} = -\frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \left[\sum_{j \in \{y_i^{(j)} = y_i^{(j)}\}} p_i^{(j)} \cdot \log(1 - u_i^{(j)}) + \frac{1}{C} \sum_{j \in \{y_i^{(j)} \neq y_i^{(j)}\}} (1 - p_i^{(j)}) \cdot \log(u_i^{(j)}) \right], \quad (13)$$

where $p_i^{(j)}$ is the maximum class probability of an input sample x_i and $u_i^{(j)}$ is the associated evidential uncertainty. In Eq. (13), the first term ensures low uncertainty when the model makes accurate predictions, thereby boosting model confidence. In contrast, the second term assigns high uncertainty to incorrect predictions to prevent overconfident mispredictions. This design balances the model's confidence in its predictions with the associated uncertainty, enabling the model to better discern when its predictions are reliable. Thus, a sample with high uncertainty is more likely to be incorrect.

3.2.3. Loss function

To leverage the evidence and uncertainty in the optimization process for prediction, we replace the probabilities in Eq. (8) with evidence-based alternatives, reformulating the EDL loss as:

$$\mathcal{L}_{EDL} = -\frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \sum_{j=1}^C y_i^{(j)} \cdot \log\left(\frac{\alpha_i^{(j)}}{S_i}\right), \quad (14)$$

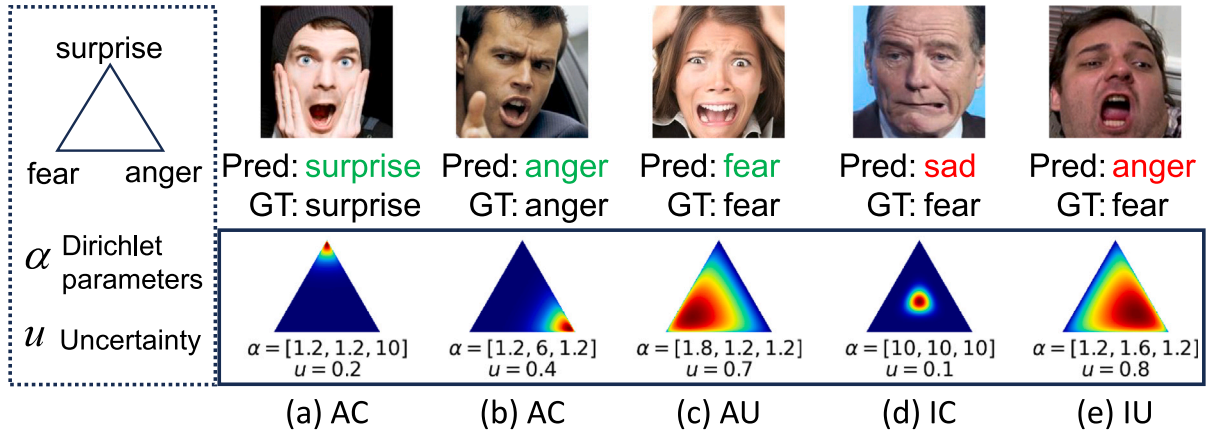


Fig. 3. Illustration of Prediction and Accuracy versus Uncertainty utility. Accurate and Certain (AC) predictions (a) and (b), and Inaccurate and Uncertain (IU) predictions (e), while the AU (c) and IC (d) cases need to be reduced.

where $\alpha_i^{(j)}$ and S_i are the parameters and strength in the Dirichlet distribution, respectively, as described by Eq. (3) to (5). Eventually, by aggregating all the aforementioned optimization objectives, we derive the total loss function as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{EDL}} + \lambda_1 \mathcal{L}_{\text{RUC}} + \lambda_2 \mathcal{L}_{\text{KD}}. \quad (15)$$

Here, λ_1 and λ_2 represent the calibration factor and the distillation factor, respectively, used to balance the total loss.

4. Experiments

In this section, we first describe the datasets and experimental settings. Then, we compare UA-FER with state-of-the-art FER methods and conduct ablation studies to analyze the impact of each module. Finally, we provide an attention visualization analysis of our UA-FER.

4.1. Datasets

We use RAF-DB [6], AffectNet [5], ExpW [7] and Multi-PLE [63], along with occlusion and pose-changes test sets as our target datasets for FER.

RAF-DB [6]: The RAF-DB dataset comprises 29,672 in-the-wild facial expression images. In line with [32], we utilize 15,339 images representing seven fundamental expressions, namely “anger”, “disgust”, “fear”, “happy”, “neutral”, “surprise”, and “sad”. Following [32], 12,271 images are allocated for training, while 3068 images are reserved for testing.

AffectNet [5]: The AffectNet stands as the largest in-the-wild FER dataset, featuring 450K annotated facial images from a total of 1M images. Following [9], we use 280K images for training and 4000 images for validation (with 500 images per category), covering eight expressions: the seven found in RAF-DB, and “contempt”.

Occlusion-RAF-DB and Occlusion-AffectNet [8]: These datasets include occluded facial images drawn from the test set of RAF-DB and the validation set of AffectNet, containing 735 and 683 images, respectively.

Pose-RAF-DB and Pose-AffectNet [8]: These datasets consist of images with facial pose variations from the RAF-DB test set and the AffectNet validation set. Specifically, they include 1248 and 1949 images with a facial pose $\geq 30^\circ$, as well as 558 and 958 images with a pose $\geq 45^\circ$, respectively.

ExpW [7]: ExpW contains seven expressions, similar to RAF-DB, and is compiled from Google Search. As per the split used in [64], 68,845 images (75%) are designated for training, 9179 (10%) for validation, and 13,769 (15%) for testing.

Multi-PLE [63]: Multi-PLE is a multi-purpose dataset in-the-lab setting, which contains six categories including “disgust”, “scream”,

“smile”, “squint”, “surprise”, and “neutral”. Following [65], we select images of 270 subjects captured at five pan angles 30° , 15° , 0° , -15° , and -30° , and each angle contains 1531 images. Consequently, there are 6124 images for training and 1531 images for testing.

For RAF-DB, ExpW, and Multi-PLE, the overall accuracy on the test set is used for measurement, while the mean class accuracy on the validation set is reported for AffectNet.

4.2. Experimental setting

For data processing, all images are resized to 224×224 . Additionally, data augmentation techniques $DA(\cdot)$, such as random flipping, random rotation, random cropping, scaling, and blurring, are employed to improve the model’s generalization. CLIP (ViT-B/16) is used as the pre-trained vision-language backbone for UA-FER, with the temperature parameter τ set to 0.07. During training on RAF-DB, ExpW, and Multi-PLE, the batch size is set to 128, and the model is trained for 30 epochs using the AdamW optimizer with an initial learning rate of 0.0001. For the large-scale dataset AffectNet, the model is trained for 6 epochs with a batch size of 256 and an initial learning rate of 0.00001. Furthermore, grid search is applied to determine all hyper-parameters. Specifically, the decoupling factor k , the calibration factor λ_1 , the distillation factor λ_2 to are set to 16, 1, and 1, respectively across all datasets. The global factor γ is set to 0.3 for AffectNet and 0.1 for the remaining datasets. All experiments are performed on a single NVIDIA-RTX-3090 GPU with 24 GB of memory and Pytorch 1.13.1.

4.3. Comparison with state-of-the-art

In this section, we compare UA-FER with current state-of-the-art (SOTA) methods on four FER benchmarks, including in-the-wild datasets RAF-DB, AffectNet, and ExpW, and in-the-lab dataset Multi-PLE, as well as occlusion and pose-changes test sets to demonstrate the superiority of our methods.

4.3.1. Comparison on RAF-DB and AffectNet

Table 1 compares our methods with existing SOTA methods on the RAF-DB and AffectNet datasets. Both datasets evaluate all methods on seven emotions, while “8” denotes an additional “contempt” expression. First, as shown in the upper part of the table, Local attention-based methods like RAN [8], SCN [32], DMUE [10], SOFT [67], AMP-Net [13], MRAN [15], and FG-AFR [14] employ ResNet to extract discriminative local features. For instance, MRAN [15] captures multi-level relationships among local regions through a multi-relation aware network. FG-AGR [14] extracts fine-grained local features using an adaptive graph association reasoning module, achieving strong performance with 90.81% on RAF-DB and 64.91% on AffectNet. Second,

Table 1
Comparison (%) with state-of-the-art methods on RAF-DB and AffectNet-7/8^a datasets.

Methods	Year	Vision backbone	RAF-DB	AffectNet-7/8 ^a
RAN [8]	TIP20	ResNet-18	86.90	59.50/52.97 ^a
SCN [32]	CVPR20	ResNet-18	88.14	60.23 ^a
DMUE [10]	CVPR21	ResNet-18	89.42	63.11
FDRL [40]	CVPR21	ResNet-18	89.72	—
EfficientFace [66]	AAAI21	ResNet-50	88.36	63.70/59.89 ^a
Meta-Face2Exp [12]	CVPR22	ResNet-50	88.54	64.23
SOFT [67]	ECCV22	ResNet-18	90.42	66.13/62.69 ^a
EAC [68]	ECCV22	ResNet-18	89.99	65.32
AMP-Net [13]	TCSVT22	ResNet-34	89.25	64.54/61.74 ^a
MRAN [15]	TCSVT23	ResNet-18	90.03	66.31/62.48 ^a
FG-AGR [14]	TCSVT23	ResNet-18	90.81	64.91/60.69 ^a
MViT [17]	arXiv21	DEiT-S/16	88.62	64.57/61.40 ^a
VTFF [16]	TAC21	R18+ViT-B	88.14	64.80/61.85 ^a
TransFER [9]	ICCV21	IR50+ViT-S	90.91	66.23
APViT [18]	TAC22	IR50+ViT-S	91.98	66.91
CLIPER [24]	ICME24	ViT-B/16	91.61	66.29/61.98 ^a
UA-FER (Ours)	2024	ViT-B/16	92.59	66.95/62.42^a

^a Denotes 8 expressions with “contempt” category.

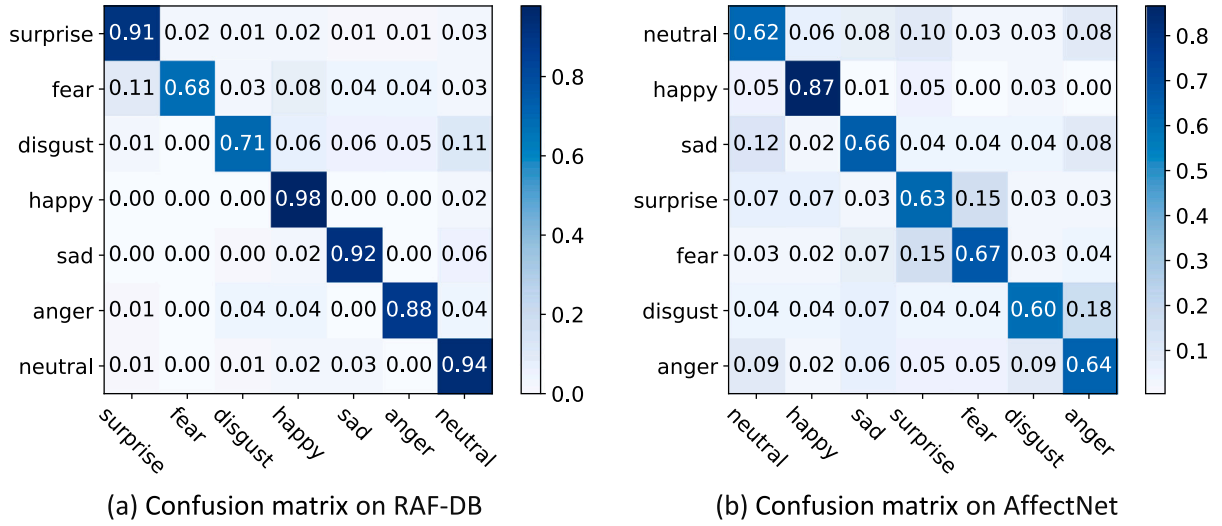


Fig. 4. Confusion matrices of the proposed UA-FER on (a) RAF-DB and (b) AffectNet.

Global perception-based methods including MViT [17], VTFF [16], TransFER [9], and APViT [18] use vision transformer for global facial modeling. For example, TransFER [9] focuses on discriminative and diverse local patches through Multi-Attention Dropping technology. APViT [18] incorporates attentive patch pooling and attentive token pooling to emphasize facial-related features, achieving the highest accuracy on RAF-DB and AffectNet with 91.98% and 66.91%, respectively.

Different from the above methods, we incorporate textual information to capture complex expression details from both visual and semantic perspectives. Additionally, it is worth noting that CLIPER [24] also conducts FER using a pre-trained CLIP, achieving performance scores of 91.61% and 66.29%. However, it relies solely on deterministic predictions, failing to capture uncertainty in image-text pairs and overlooking fine-grained localized facial features. In contrast, our UA-FER achieves accuracy of 92.59%, 66.95%, and 62.42% on RAF-DB, AffectNet-7, and AffectNet-8[†], respectively, surpassing 1.34%, 0.66%, and 0.44% over CLIPER. This demonstrates that our method effectively captures both local and global facial representations from visual and semantic perspectives, while also modeling and correcting visual-textual relationships.

In addition, Fig. 4 presents the confusion matrix for UA-FER on the RAF-DB and AffectNet datasets. For the RAF-DB dataset (Fig. 4(a)),

our method achieves promising results across most classes, particularly 91%, 98%, 92%, 88%, and 94% for “surprise”, “happy”, “sadness”, “anger”, and “neutral”, respectively. However, the unbalanced data distribution in RAF-DB, where “fear” and “disgust” each constitute less than 10% of the data, leads to lower performance due to higher confusion rates. As shown in Fig. 4(b), the AffectNet dataset, with its large number of unconstrained facial expression images, presents greater recognition difficulty compared to RAF-DB due to its higher complexity. The figure shows that our method achieves the highest recognition rate of 87% for the “happy” expression. In comparison, the other six expression classes display similar performance (60%–67%), with “surprise”, “fear”, and “disgust” being particularly challenging to recognize due to frequent confusion.

4.3.2. Comparison on occlusion and pose-changes datasets

To further evaluate the performance of our UA-FER in more complex scenarios, we conduct experiments on occlusion and pose change datasets, including Occlusion-RAF-DB, Occlusion-AffectNet, Pose-RAF-DB, and Pose-AffectNet [8]. Tables 2 and 2 show the comparison of our method with current SOTA methods, such as RAN [8], EfficientFace [66], MA-Net [69], VTFF [16], AMP-Net [13], and MRAN [15]. According to the experimental results in Table 2, our UA-FER achieves

Table 2
Comparison (%) with state-of-the-art methods on occlusion and pose-changes datasets.

Methods	Year	Occlusion	Pose $\geq 30^\circ$	Pose $\geq 45^\circ$
RAN [8]	TIP20	82.73	86.74	85.20
EfficientFace [66]	AAAI21	83.24	88.13	86.92
MA-Net [69]	TIP21	83.65	87.89	87.99
VTFF [16]	TAC21	83.95	87.97	88.35
AMP-Net [13]	TCSVT22	85.28	89.75	89.25
MRAN [15]	TCSVT23	86.94	90.38	89.78
UA-FER (Ours)	2024	89.93	91.56	91.40
(a) The results on Occlusion-RAF-DB and Pose-RAF-DB.				
Methods	Year	Occlusion	Pose $\geq 30^\circ$	Pose $\geq 45^\circ$
RAN [8]	TIP20	58.50	53.90	53.19
EfficientFace [66]	AAAI21	59.88	57.36	56.87
MA-Net [69]	TIP21	59.59	57.51	57.48
VTFF [16]	TAC21	62.98	60.61	61.00
AMP-Net [13]	TCSVT22	64.27	61.37	61.16
MRAN [15]	TCSVT23	63.54	61.42	62.05
UA-FER (Ours)	2024	66.95	63.85	63.29
(b) The results on Occlusion-AffectNet and Pose-AffectNet.				

Table 3
Comparison (%) with state-of-the-art methods on ExpW dataset.

Methods	Year	ExpW
DCN [7]	IJCV18	65.06
DCN+AP [7]	IJCV18	70.06
FAUCNN [70]	Access19	72.84
SchiNet [64]	TAC19	73.10
PAT-Res101 [71]	TAC22	69.29
PAT-Res101+attr [71]	TAC22	72.93
APViT [18]	TAC22	73.48
UA-FER (Ours)	2024	74.55

recognition accuracy of 89.93%, 91.56%, and 91.40% on Occlusion-RAFDB, Pose-RAF-DB (Pose $\geq 30^\circ$), and Pose-RAF-DB (Pose $\geq 45^\circ$), which is an improvement of 2.99%, 1.18%, and 1.62%, respectively, compared to the best current method MRAN. As shown in Table 2, the proposed method still achieves the highest accuracy on large-scale occlusion and pose-change datasets, surpassing MRAN by 3.41%, 2.43%, and 1.24% on Occlusion-AffectNet, Pose-AffectNet (Pose $\geq 30^\circ$), and Pose-AffectNet (Pose $\geq 45^\circ$), respectively. These results demonstrate that the model's robustness in realistic and complex scenarios (e.g., occlusion and pose changes) can be significantly enhanced by introducing textual information and correcting the uncertainty with the corresponding visual part.

4.3.3. Comparison on ExpW

ExpW is a challenging dataset, known for its complex backgrounds and multi-instance characteristics. Table 3 presents the comparison of our proposed method with other SOTA methods on the ExpW dataset. FAUCNN [70] oversamples images with similar emotions by utilizing action unit (AU) features. SchiNet [64] merges multiple datasets into a large-scale facial dataset for training, achieving an accuracy of 73.10%. APViT [18] combines a multi-branch local CNN with ViT to extract attention information from individual local patches, achieving an accuracy of 73.48%. Using only the target dataset, our UA-FER achieves an accuracy of 74.55%, surpassing existing methods. We attribute this to the model's ability to effectively mine and correct uncertainties in vision-language relationships by leveraging textual information and uncertainty modeling, significantly enhancing the model's recognition accuracy and robustness.

4.3.4. Comparison on Multi-PLF

We compare the proposed UA-FER with several SOTA methods on Multi-PLF in Table 4. Results across all poses are summarized in column 2~6 and the mean accuracy is reported in the last column. We evaluate the performance of kNN, GMLPP, MvDA, DS-GPLVM reported

in Eleftheriadis et al. [72], and deep learning methods CG-FER [73], UIGA-FER [74], JESRL-FER [65] and SSA-Net [75] for pose-invariant FER. CG-FER enhances the model by generating facial images with arbitrary expressions and poses, while the JESRL-FER achieves remarkable performance by extending the training samples based on a GAN network. As we can see, our model achieves 93.75% mean FER accuracy and surpasses these methods. Besides, while most algorithms struggle to achieve satisfactory performance in the frontal view, the proposed method successfully improves it, achieving a significant performance increase from 1.5% to 26.89%. However, we observe that at wider pose angles of 15° and 30° , our model performs slightly worse than the optimal methods. This may be because, while our UA-FER integrates uncertainty mechanisms to learn pose-invariant features, the complex non-linear variations at larger angles still require specialized approaches, such as explicit 3D transformation modeling or enhancing the feature extractor with angle-specific modules. Overall, the proposed method achieves the most competitive performance by learning discriminative and generalizable facial representations from visual and linguistic perspectives.

4.4. Ablation study

In this section, we conduct ablation studies to evaluate the contributions of the key modules in UA-FER: Multi-granularity Feature Decoupling (MFD) and Relation Uncertainty Calibration (RUC), as well as important hyper-parameters. All the results are reported on RAF-DB and AffectNet.

4.4.1. Effectiveness of the proposed modules

To evaluate the effectiveness of our proposed modules, we perform ablation studies on MFD and RUC on RAF-DB and AffectNet datasets. Specifically, MFD focuses on capturing expression-related discriminative features using pre-trained CLIP models guided by textual

Table 4

Comparison (%) with state-of-the-art methods on Multi-PLE. The highest accuracy for each pose is highlighted in bold.

Methods	Year	Poses					Average (%)
		-30°	-15°	0°	15°	30°	
kNN [72]	TIP14	80.88	81.74	68.36	75.03	74.78	76.15
GMLPP [72]	TIP14	91.86	94.13	78.16	87.22	87.36	87.74
MvDA [72]	TIP14	92.49	94.22	77.51	87.10	87.89	87.84
DS-GPLVM [72]	TIP14	93.55	96.96	82.42	89.97	90.11	90.60
CG-FER [73]	CVPR18	90.76	94.72	89.11	93.09	91.30	91.80
UIGA-FER [74]	VCIP19	93.07	93.77	92.83	92.21	95.13	93.40
JESRL-FER [65]	TCSVT21	93.83	95.13	93.16	92.78	93.39	93.66
SSA-Net [75]	PR23	93.59	93.38	93.75	95.59	93.38	93.54
UA-FER (Ours)	2024	95.50	95.25	95.25	92.23	92.41	93.75

Table 5

Evaluation (%) of the key modules in UA-FER. “w/o.” means “without”.

Modules	RAF-DB	AffectNet
baseline	90.58	65.89
MFD (w/o. $\mathcal{L}_{KD}$)	91.16	66.09
MFD	91.48	66.23
MFD+RUC ($\mathcal{L}_{EDL}$)	91.85	66.57
MFD+RUC ($\mathcal{L}_{EDL} + \mathcal{L}_{RUC}$)	92.59	66.95

information, while RUC aims to improve model robustness by correcting uncertainty in visual-textual relationships. The baseline model (Line 1) represents a fine-tuned CLIP model where only the last transformer block of the image encoder is trained, while the rest of the image and text encoder layers remain frozen. As shown in Table 5, both modules significantly improve the performance compared to the baseline strategy. First, MFD (Line 3) surpasses the baseline (Line 1) by 0.90% and 0.34% on the RAF-DB and AffectNet datasets, respectively, and improves accuracy by 0.32% and 0.14% compared to the model without knowledge distillation (Line 2). These results indicate that MFD effectively captures expression-related facial representations while retaining universal knowledge from pre-trained CLIP model. Second, introducing an uncertainty-based prediction strategy (Line 3) further improves accuracy by 0.37% and 0.34%, compared to softmax-based prediction (Line 2), highlighting the importance of uncertainty modeling. Finally, the full model (Line 5), incorporating all modules, outperforms all other variants across all experiments, demonstrating the effectiveness of uncertainty correction in enhancing model robustness.

4.4.2. Effectiveness of text prompts

As mentioned in CoOp [76], for pre-trained vision-language models, the text input (*i.e.*, the prompt) is crucial for downstream tasks. To verify the effectiveness of different prompting strategies, we conduct ablation studies on various hand-crafted prompts in MFD. As shown in Table 6, subtle changes in text prompts significantly impact model performance. Interestingly, for FER, optimal performance is achieved by using only category labels (*e.g.*, $[class_c]$) as prompts, which outperforms the generic cue “a photo of $[class_c]$ ” in the default setting. We hypothesize that this is due to the highly ambiguous nature of facial expressions, where additional textual information may introduce redundancy and interfere with the model’s discriminative ability. In contrast, simple category labels (*e.g.*, “happy”) provide concise and explicit semantic information, which more effectively aids the model in distinguishing different expression categories.

4.4.3. Effectiveness of hyper-parameters

We conduct ablation experiments to evaluate the impact of various hyper-parameters, including the global factor γ , decoupling factor k , as illustrated in Fig. 5, and the calibration factor λ_1 along with the distillation factor λ_2 , as shown in Fig. 6.

To evaluate the impact of the Emotion-aware Adapter, we perform ablation studies on two critical parameters in Eq. (8): (1) The **global**

Table 6

Evaluation of (%) different text prompts.

Text prompts	RAF-DB	AffectNet
$[class_c]$	92.59	66.95
a photo of $[class_c]$.	92.43	66.51
a photo of a $[class_c]$ face.	91.76	66.31
a $[class_c]$ facial expression.	92.01	66.04
a person is $[class_c]$ in the photo.	92.12	66.28
this is a $[class_c]$ facial expression image.	91.77	65.93

factor γ : As can be seen in Fig. 5(a), smaller γ degrades model performance, indicating that a reduced global factor weakens the model’s global modeling capacity. Conversely, a larger γ causes the model to overlook critical local facial regions, resulting in suboptimal results. Notably, relying solely on global ($\gamma = 1$) or local ($\gamma = 0$) representations yields lower accuracy, underscoring the importance of integrating both. Therefore, we set γ to 0.3 for AffectNet and 0.1 for the other datasets. (2) The **decoupling factor k** : As depicted in Fig. 5(b), smaller k values are more susceptible to noise in local representations, while larger k values introduce extraneous representations that impair performance. Accordingly, we choose $k = 16$ as the default setting for its optimal performance.

In addition, we investigate the impact of the **calibration factor λ_1** and the **distillation factor λ_2** on the total loss function in Eq. (15), as illustrated in Fig. 6. It shows that both excessively large and small weights negatively affect the optimization of the total loss, resulting in reduced recognition performance. Specifically, a small λ_1 fails to adequately correct the uncertainty in the visual-textual relationship, leading to numerous false predictions. Conversely, a large λ_1 may lead to overcorrection, disrupting the model’s focus on accurate image-text pairs. Similarly, a small λ_2 risks causing the model to forget the generalized knowledge from the pre-trained CLIP, while a large λ_2 may cause the model to overlook expression-related discriminative representations. Consequently, we set λ_1 and λ_2 to 1 by default.

4.5. computational complexity

We further evaluate the computational complexity of UA-FER by comparing its floating point operations (FLOPs) and parameter count (Params) with other methods on the AffectNet dataset. Notably, the FLOPs and Params for MA-Net, VTFF, AMP-Net, and MRAN are reported from Chen et al. [15], while those for MViT, TransFER, and APViT are from Xue et al. [18]. AMP-Net and MRAN are CNN-based methods, while VTFF, MViT, TransFER, and APViT are based on the Transformer. Compared to TransFER and APViT, our UA-FER achieves the highest performance (66.95%), requiring only 73.66% and 88.95% of the FLOPs of them. However, UA-FER exhibits higher FLOPs than VTFF and MViT, primarily due to the multi-head self-attention computations introduced by the additional Transformer blocks in the CLIP-like ViT-Base architecture. Despite the increased computational complexity, our UA-FER significantly surpasses VTFF and MViT with performance gains of 2.15% and 2.38%, respectively. Additionally, Table 7 shows

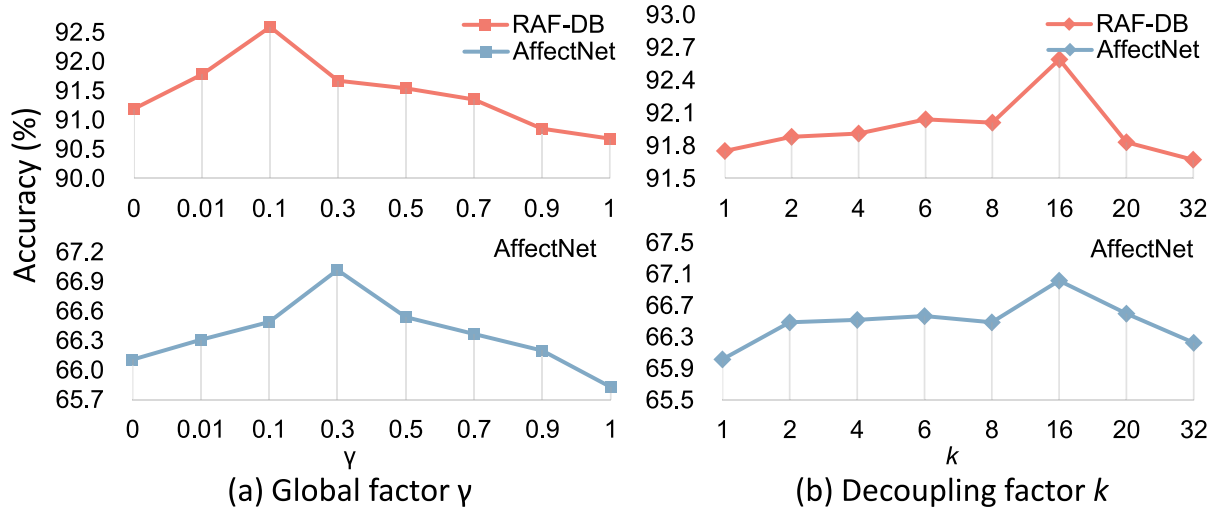


Fig. 5. Influence of global factor γ and decoupling factor k in Eq. (8) on RAF-DB and AffectNet datasets.

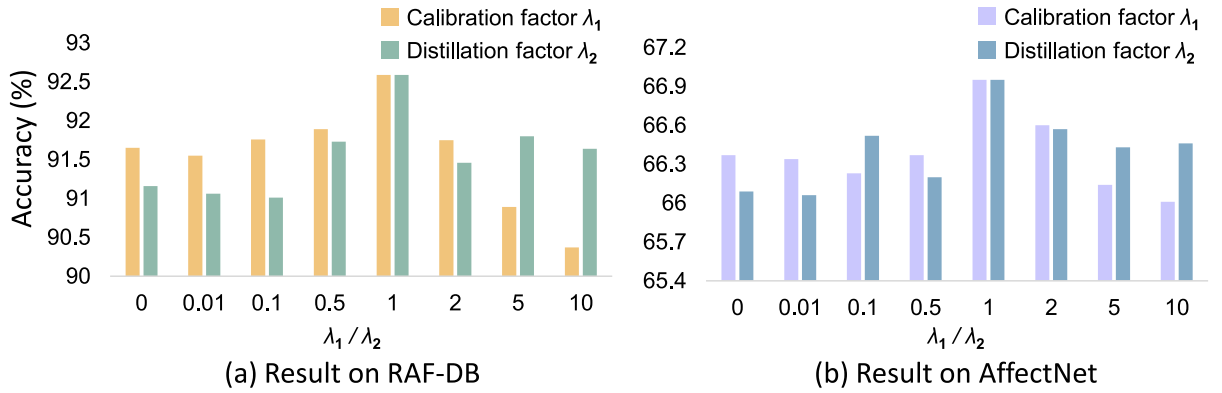


Fig. 6. Influence of calibration factor λ_1 and distillation factor λ_2 in Eq. (15) on RAF-DB and AffectNet datasets.

Table 7

The number of parameters (M) and flops (G) obtained by different methods on AffectNet dataset.

Methods	FLOPs	Params	AffectNet
AMPNet [13]	4.73 G	105.67 M	64.54
MRAN [15]	3.89 G	60.52 M	66.31
VTFF [16]	6.08 G	80.10 M	64.80
MViT [17]	5.95 G	–	64.57
TransFER [9]	15.30 G	–	66.23
APViT [18]	12.67 G	–	66.91
UA-FER (Ours)	11.27 G	28.74 M	66.95

that our UA-FER achieves the highest performance (66.95%), with 52% fewer trainable parameters than MRAN (66.31%). This demonstrates that UA-FER effectively reduces parameter usage while learning emotion-specific representations and preserving universal knowledge from the pre-trained CLIP model.

4.6. Attention visualization

To visually demonstrate the effectiveness of UA-FER, we use ScoreCAM [77] to visualize several images randomly selected from the RAF-DB and AffectNet datasets. As illustrated in Fig. 7, lighter colors indicate stronger ScoreCAM activations. Firstly, we observe that the pre-trained CLIP model (Row 2) tends to focus on irrelevant or distracting regions (e.g., the edge of the image) while neglecting expression-related facial muscle regions (e.g., Columns 1, 3, 5 in Row 2 of subfigure (a),

and Columns 1, 7 in Row 2 of subfigure (b)). We attribute this to the pre-trained CLIP model's inclination to focus on global object-level representations, overlooking subtle local representations crucial for FER.

In contrast, our proposed MFD module (Row 3) effectively focuses on expression-related regions. This is because the MFD module fine-tunes the pre-trained CLIP model specifically for the FER task, enabling it to capture expression-related discriminative representations more effectively while retaining as much generalized knowledge from the pre-trained CLIP model as possible. Besides, in complex environments (e.g., under- or over-illumination), the pre-trained CLIP model often fails to focus on facial regions, whereas our model accurately captures facial regions associated with emotional expressions (as shown in Column 2 of subfigures (a) and (b)). Furthermore, our proposed RUC module (Row 4) demonstrates significant advantages in handling ambiguous visual-textual relationships within MFD. When ambiguous relationships arise between different expression images and emotional text due to their similar appearances (e.g., frowns in Columns 4 and 5 of subfigures (a) and (b)), the RUC module effectively aids the model in distinguishing these expressions by correcting the uncertainty, eliminating redundant representations, and focusing precisely on the remaining discriminative representations. Additionally, the proposed UA-FER (i.e., MFD+RUC in Row 4) excels in further emphasizing key discriminative regions, as seen in Columns 3, 6, and 7 of subfigures (a) and (b). These results suggest that our UA-FER model effectively develops an emotion-aware feature extractor capable of accurately modeling the complex relationship between expression images and emotional text, thereby significantly enhancing recognition performance.

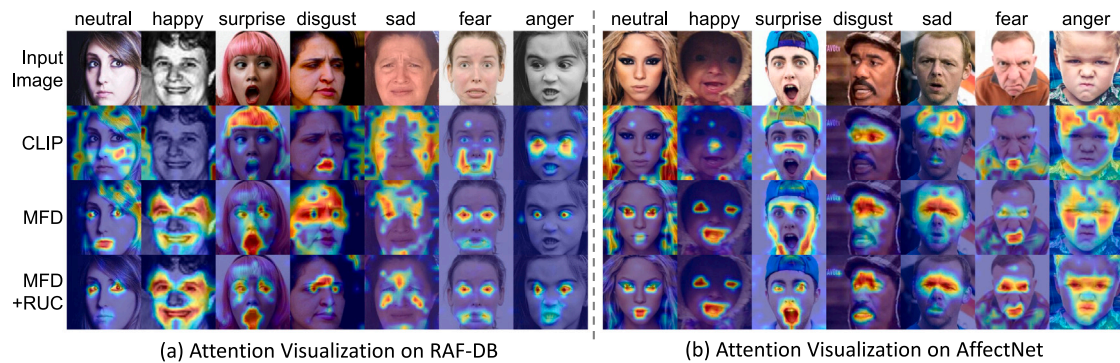


Fig. 7. Attention visualization [77] on (a) RAF-DB and (b) AffectNet datasets.

5. Conclusion

In this paper, we propose a uncertainty-aware representation learning framework for FER, which integrates pre-trained CLIP model with evidential deep learning theory to enhance recognition accuracy and robustness. We propose a Multi-granularity Feature Decoupling module to identify multi-grained discriminative regions, which decouples global and local facial representations by leveraging image-text affinity. Furthermore, knowledge distillation is employed to prevent our model from forgetting the pre-trained universal knowledge. Additionally, we introduce the Relation Uncertainty Calibration module to address misjudgments in uncertain visual-textual relationships. We conduct extensive experiments on both in-the-wild and in-the-lab FER datasets, and the results demonstrate the effectiveness of our method. Moreover, we present visualization studies to illustrate how our method focuses on critical facial regions when identifying expressions. Future work will focus on integrating pose-aware feature augmentation and multi-view learning strategies to address large pose variations in facial images. We also plan to deploy our model to more realistic and challenging FER scenarios, including zero-shot and cross-domain settings.

CRedit authorship contribution statement

Haoliang Zhou: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Data curation, Conceptualization. **Shucheng Huang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yuqiao Xu:** Writing – review & editing, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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